

Evolution of GIS-based rural electrification planning models and an application of OnSSET in Nigeria

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ABSTRACT

This study carries out an in-dept overview of GIS-based rural electrification planning models with a trace to their historical evolutions. The study focuses on web-based and desktop-based models that have been developed and used extensively in rural electrification planning in recent years. Specifically, four of such models are considered: SOLARGIS, Network Planner, the Reference Electrification Model (REM), and Open Source Spatial Electrification Tool (OnSSET). We compare the performance of these models and conclude that OnSSET has better capabilities than the other models. Specifically, we find OnSSET very useful because it is open source. We demonstrate the application of OnSSET in a state in Nigeria, i.e. Kaduna State. The state is selected because of data availability. The result of the study shows that mini-grid solar PV is the cost-effective technology option for most unelectrified communities, followed by Standalone PV. Furthermore, the cost of achieving universal electricity access in the state by 2030 is estimated at US\$505.08million while an additional generation capacity of 252MW will be required to achieve universal electricity coverage by 2030. The study concludes that the financial requirement seems to be beyond the capacity of Kaduna State given the prevailing economic realities in Nigeria and recommends that pro-poor public-private partnership strategies be considered for electrification projects in the state.

Background

Despite the importance of modern energy services, especially electricity, about 771 million persons do not have access to electricity [1]. Increasing access to modern energy services has become one of the global development challenges and is one of the Sustainable Development Goals – specifically, Goal 7. In line with this, access to electricity has received significant attention from researchers and policymakers in recent years. The provision of electricity to the millions of persons that currently lack access usually requires systematic planning to obtain the cost-effective option between extending the existing grid and using off-grid electrification. Such planning takes cognizance of the availability of energy resources in each unelectrified community and also incorporates spatial diversity (such as diversity in demographic attributes, terrain types, etc.) [2]. The traditional form of rural electrification planning requires a large set of data that are crucial for evaluating, planning, implementing, and monitoring the provision of energy access services to un-electrified areas. Some of these data are location-specific, very expensive, and laborious to obtain [3–5].

In recent times, researchers have applied geographic information system (GIS) to overcome some of these data-related challenges [6]. Consequently, several models¹ have been developed on GIS-based rural electrification planning and applied in different countries at national and sub-national levels [4,6–10]. Some of these models are Logiciel d'Aide à la Planification d'Électrification Rurale (LAPER), SOLARGIS,² Network Planner, Reference Electrification Model (REM), and Open Source Spatial Electrification Tool (OnSSET). Even though these models all employ GIS to aid rural electrification planning, they differ in terms of their data requirement, methodology, cost, accessibility, amongst others. The objective of this study is to examine the evolution of GIS-based rural electrification planning tools, provide an in-dept overview of these tools, and to demonstrate the application of OnSSET in Nigeria. For the OnSSET analysis, we use 2018 as the base year, 2025 as the step year, and 2030 as the final year. There have been previous studies that have reviewed energy planning models [3,11–20]. There are also recent stud-

¹ “Models” and “Tools” are used interchangeably in this study.

² It is important to state that this application is NOT the same as the new generation web service that provides web-based high-resolution solar data, other meteorological data, PV simulation software, solar maps, GIS database, and PV monitoring services for solar energy project developers (<http://solargis.info> or <http://solargis.com/>, visited on August 4, 2019).

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ies that have reviewed commonly-used GIS-based rural electrification planning tools [21–23]. This study contributes to this body of knowledge in the sense that it attempts to trace how these models have evolved and compares them using several parameters.

This study defines GIS-based rural electrification planning tools as autonomous desktop or web-based applications that may be used to obtain the least-cost electrification options in a country or sub-national region. The essence of the GIS is to carry out electrification planning at a spatially disaggregated level. Based on this definition, tools such as RETScreen and Hybrid Optimization of Multiple Energy Resources (HOMER), Long-range Energy Alternatives Planning system (LEAP) and Open Source Energy Modelling System (OseMOSYS) are not considered in our study because the purpose of these models is different and other previous studies have already reviewed these models [3,16,24–26].

Following this Background, we present an overview of the application of GIS to rural electrification planning in Section 2 and demonstrate the application of OnSSET in Section 3 while Section 4 provides a discussion and provides a discussion and concluding remarks.

An overview of applications of GIS in rural electrification planning

Historical evolution of models

Studies in the 1990s

One of the earliest works that have applied GIS in rural electrification planning was done by [27]. The study built on some earlier works which compared the cost of grid-extension with decentralized options for rural electrification [28–30], and examined the territorial competitiveness of standalone solar photovoltaics using paper maps. Before this period, a consortium of research centres within the European Union's Joint Opportunities for Unconventional or Longer-term Energies (JOULE) program had developed a computer programme called SOLARGIS. This model is to help in assessing the market potential for renewable energy technologies for unserved regions in Europe. SOLARGIS was tested in three developed countries (Andalucia, Spain; Sicily, Italy; Crete, Greece) and three developing countries (Tunisia; India; Cabo Verde) [31,32]. Similarly, Muselli *et al* [33] applied GIS to realize a plan for integrating various renewable energy technologies in remote areas in Corsica, an island located in the southeast of France.

Studies after the turn of the millennium

By the turn of the millennium, and in the light of the Millennium Development Goals (MDGs), studies on the provision of electricity access to underserved communities increased several folds. The increase in focus on rural electrification was in part because countries elected to reduce their dependence on fossil fuels to reduce greenhouse gas emissions, promote energy security, and by extension increase the use of renewable energy technologies. Earlier studies had already demonstrated the potential of renewable energy technologies for electrifying isolated and underserved communities with relatively low demand. Fronius & Gratton [34] presents a study using rural electrification planning software tool called LAPER (i.e. Logiciel d'Aide à la Planification d'Électrification Rurale) which searches for optimal electrification solutions in underserved areas using a step-by-step replacement of alternative energy sources while taking into account geographical constraints. LAPER also considers other factors that may influence decision-making vis-à-vis rural electrification such as political considerations, development impact, and the availability of financial resources. Similarly, Soler *et al* [35] applied LAPER to investigate the potential for large-scale deployment of renewable energy systems in Morocco. Given that the terrain in rural areas is usually not uniform and differences in terrain may increase the cost of grid-extension, Luchmaya *et al* [36] applied a GIS-based rural electrification planning model that incorporates cost penalties for differences in terrain when carrying out grid-extension.

Amador & Dominguez [37] and Amador & Dominguez [38] improved on the methodology of SOLARGIS and demonstrated the improvements in the municipality of Lorca (Murcia, Spain). Specifically, Amador & Dominguez [37] improved on the study by Ariza-Lopez *et al* [27] by using vector maps of medium voltage lines and housing as well as raster maps of solar and wind resources. The study makes improvements in the computation of the levelized cost for each electrification option. Amador & Dominguez [38] augmented the approach by carrying out a spatial sensitivity analysis to determine the parameters that have a great influence on the results.

A substantial effort is undertaken by Kaijuka [39] who investigated the patterns of demand and priority areas for investment for rural electrification planning in Uganda. The study used a GIS approach and noted that by creating a demand-side scenario, electricity can then be supplied to targeted areas by way of allocating an energy benefit point system to priority sectors (education and health) based on local conditions and available energy resources in each area. However, this study did not consider other energy supply systems and does not suggest an optimal way to provide electricity to the identified priority areas. On their part, Tiba *et al* [40] developed a GIS-based decision support tool for renewable energy management and planning in rural areas called GISA SOL 1.0. The tool allows planning of a sizeable addition of renewable energy technologies and the management of the already installed systems within communities. Different criteria were considered to identify the most suitable location for installing new energy systems. These criteria include solar resource availability, human development index, rural electrification index, income per capita, proximity to a transmission line, the existence of potable water, wind resources, restriction of soil use and others. The study considers mainly the implementation of solar system, wind power, and biomass technologies, thus overlooking the potential penetration of other technologies (such as grid-extension or mini-hydropower and diesel generators) to provide electricity to unserved areas. The Improving Economic and Social Impact of Rural Electrification (IMPROVES-RE) Programme in Burkina Faso was another novel attempt to use GIS to support rural electrification projects and used a web-based platform [41]. This was followed by yet another tool called GeoSIMS which is developed by the same organization (i.e. Innovation Energie Développement) but operates under a proprietary platform called Manifold. GeoSIMS arranges potential electrification sites according to their development potentials, selects a few of them to be "Development Poles" and clusters the communities around them.

A distinct strand of GIS-based rural electrification planning model was initiated by Parshall *et al* [42] who incorporated geospatial information on population settlement patterns into an electricity planning model to identify the least-cost technology option for providing electricity to different demand centers in Kenya where urban and rural electrification rate was 30% and less than 10%. The unique feature of the study was the application of the Kruskal's minimum spanning tree algorithm to select a set of demand areas that will minimize the total cost of grid extension. This approach was also used by Sanoh *et al* [43] in Senegal. However, these studies do not incorporate the general costing associated with grid extension and other renewable energy solutions. Additionally, the algorithm is biased towards grid-extension. Further development and refinement of this approach was codified into a web-based model for rural electrification planning called Network Planner. The tool was formally introduced by Modi *et al* [44] who applied it in Liberia. The notable advantage of the Network Planner is that it was free, web-based, integrated with Google Maps, and relatively easy to use. Consequently, the tool has been applied in Ghana [7]; and Nigeria [2,45,46]. Abdul-Salam & Phimister [47] adopts an approach similar to that of Network Planner, combines it with hierarchical lexicographic programming to reflect the politico-economic dimensions of community preference to grid-electricity, and applies the model in Ghana.

Similarly, Mentis *et al* [4] introduced another web-based and GIS-based electrification planning tool called Open Source Spatial Electrification Tool (OnSSET). The tool is used to inform electrification planning

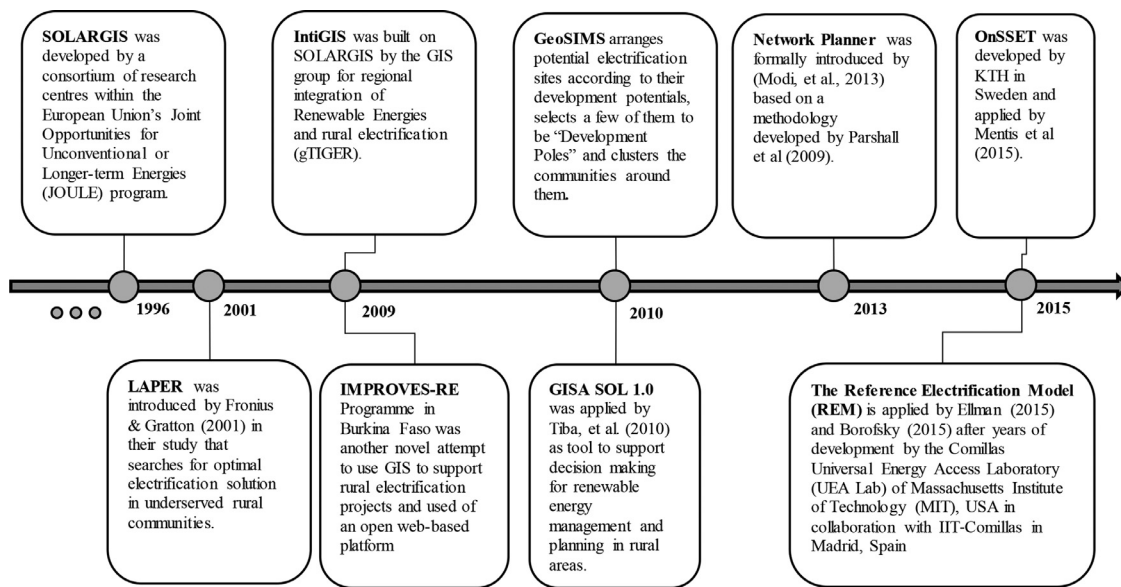


Fig. 1. Timeline for the application of GIS-based rural electrification planning models.

and strategies for achieving energy access targets at different tiers of access. The study applied the tool in Nigeria to identify the optimal mix of electrification options, ranging from grid extensions to mini-grid and off-grid solutions. Mentis *et al* [4] used several variables such as population density, transmission line (existing and planned) power plants, economic activities, tariffs for grid-based electricity, technology costs for mini-grid and off-grid systems, and fuel costs for consumers. To further advance the application of the tool, Mentis *et al* [6] demonstrates the usefulness of the OnSSET model in providing insights into electrification planning by applying it in Sub-Saharan Africa (country-by-country basis) to determine the least-cost options amongst grid extension, mini-grid (wind, hydro, diesel sets) and stand-alone systems for providing electricity infrastructure for rural electricity access using local relevant information for rural, peri-urban, and urban areas. Other studies that used the OnSSET tools include [9,10,48–51].

Martinez-Cesena *et al* [52] introduced a GIS-based framework that integrates mobile phone data analysis, socioeconomic indicators, geospatial data analysis, and state-of-the-art energy infrastructure engineering techniques to assess the viability of centralized and decentralized electrification options (grid-extension, diesel engine-based micro grids and stand-alone solar photovoltaic systems) for rural areas in Senegal. Cader *et al* [53] introduced an un-named tool for developing techno-economically optimized renewable electrification strategies for Sub-Saharan Africa and applied the tool in Nigeria. The method adopted by the tool seems to follow the approach of Network Planner. Another model that has received considerable attention in recent times is the Reference Electrification Model³ [54–56]. The timeline for the evolutions of these GIS-based rural electrification planning models is presented in Figure 1

Apart from tools that employ GIS in determining the least-cost option for electrifying underserved areas, there are other GIS-based tools that have been developed to support the visualization of data related to renewable energy technologies which may aid rural electrification planning. These include ECOWREX,⁴ RE2nAF⁵, etc. There are other GIS tools that have been developed to support in deciding the appropriateness of rural areas for renewable energy technologies [57].

Comparison of GIS-based rural electrification models

This section elaborates on some of the widely used tools, i.e. SOLARGIS, Network Planner, OnSSET, and REM. It should be noted that these tools only support rural electrification planning. In practical terms, there are also political, administrative, economic, regulatory, legal, social, and cultural factors, amongst others, which play important roles in decision making.

SOLARGIS

One of the first applications that integrated GIS into rural electrification planning is SOLARGIS which was developed by a consortium of research centres within the European Union's Joint Opportunities for Unconventional or Longer-term Energies (JOULE) program [58]. Specifically, the SOLARGIS consortium was made up of: ARMINES, France; CIEMAT, Spain; CONPHOEBUS, Italy; CRES, Greece; INESC, Portugal; NMRC, Ireland; and RAL, United Kingdom⁶ [31,32]. The primary objective of SOLARGIS was to demonstrate the viability of applying GIS in the identification of the right technologies for distribution network expansion in any region and assessing the economic feasibility of each technology.

IntiGIS is another software built on SOLARGIS by the GIS group for regional integration of Renewable Energies and rural electrification (gTIGER) [58,59]. IntiGIS is a dynamic, flexible, interoperable and modular tool that incorporates new renewable energy technologies and also simplifies the required parameters. The tool has been applied in some Latin American countries like Cuba, Guatemala and Columbia [58]. In the late 2012, the Economic Community of West African States (ECOWAS) Centre for Renewable Energy and Energy Efficiency (ECREEE) launched the ECOWREX project to help decision makers, project developers, investors and other stakeholders with tailored information and planning strategies for improved renewable energy penetration in the ECOWAS region. The IntiGIS 2.0 – an electricity access planning tool which builds on IntiGIS was developed to support

³ <http://universalaccess.mit.edu/rem-web-demo/> (visited on August 5, 2019).

⁴ <http://www.ecowrex.org/page/about> (visited on August 4, 2019).

⁵ <https://re.jrc.ec.europa.eu/re2naf.html> (visited on August 4, 2019).\d.

⁶ CIEMAT: Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas [Centre for Energy, Environment and Technology], Spain; CRES: Center For Renewable Energy Sources; INESC: Instituto de Engenharia de Sistemas e Computadores [Institute for Systems Engineering and Computers], Portugal; NMRC: National Microelectronics Research Centre, Ireland; RAL: The Rutherford Appleton Laboratory, United Kingdom.

the ECOWREX project. Another tool that is similar to SOLARGIS in its design and functions is the Regional Energy Planning using GIS technology (ENERGIS) which has been developed by INESC (Instituto de Engenharia de Sistemas e Computadores) of Porto in Portugal.

In terms of data requirement, the SOLARGIS model required spatial data on wind and solar energy potential in a study area. In the application of SOLARGIS in Cape Verde, Monteiro & Miranda [31] and Monteiro *et al* [32] used primary data on wind speed at 10m and developed an annual wind speed grid. Similarly, primary data on solar irradiation was obtained and used. SOLARGIS seeks to identify the right technologies for providing electricity to unserved households and communities. The model considered the leveled cost of generating electricity to meet a pre-defined demand level using two main electrification options, i.e. use of isolated systems and the extension of the existing grid. For the isolated systems, the model considered the following: (i) standalone solar PV systems with inverters and batteries (ii) standalone wind systems with inverters and batteries (iii) small individual systems, wind or solar PV with no storage, for water pumping (iv) small gasoline or diesel generators supplying isolated houses (which is used to compare with the performance of the renewable energy sources (v) hybrid wind/diesel systems to supply small towns, and (vi) diesel system to power small towns. On the other hand, for grid-extension, the model examines the cost of the existing network as well as the connecting to a secondary medium voltage (MV)/low voltage (LV) substation and the cost of the low voltage lines. In the final step, the model carries out a comparison between the different alternatives to develop a competitiveness map. Thereafter, the map that show the market potential for each renewable energy option is developed [31,32].

Network planner

The Network Planner is a decision support tool for exploring electrification options in rural communities. It has been used to compare the long-term costs of installing solar, diesel or grid distribution networks and perform sensitivity analyses on variables such as demand growth for electrification plans in Senegal, Ghana, and Nigeria. The tool was developed by the Quadracci Sustainable Engineering Lab of the Earth's Institute, Columbia University, USA. Notable works that have used this tool are [2,7,8,42,44,46]. Some advantages of Network Planner includes: (i) the tool is freely available on the web to use and it is fully integrated with Google Maps thereby taking advantage of advancement in spatial technologies; (ii) it is generally easy to use and does not require advance knowledge of programming or GIS; (iii) the tool has a good level of granularity and may be applied at local, sub-national or national levels. Furthermore, Network Planner requires data at different levels – household, community, and national, and also requires different types of data which may be grouped into five categories. The data requirement and the modelling procedure of the Network Planner is present below:

Geospatial data

Network Planner requires data on the spatial location (longitude and latitude) of demand centers. A demand centre or node may be regarded as the lowest administrative unit of a country for which data may be available. The tool also requires data on the coverage of the existing medium voltage or low voltage lines in the area where demand centers are located usually as an ESRI shapefile.

Demographic variables

The tool requires base population of the demand nodes which is the population of each demand node in the initial year in which investment for increasing electricity access will be made. Other demographic variables required are: data on population growth rate for the demand nodes; population threshold which is defined as the number below which a demand node is assumed to be rural and above which a demand node is assumed to be urban; average household size (rural and urban); the average inter-household distance measures the average distance between households in a demand node.

Economic variables

These include economic growth rate of the study area; investment horizon: the time required to increase electricity access rate to a pre-defined level; lending rate; income elasticity of electricity demand Electricity

Demand

The tool requires data on electricity demand for the demand nodes which comprises of household electricity demand; productive demand; and institutional demand.

Costs

Network Planner requires the data on the detailed cost of the three technology options that are examined: extending the existing grid; constructing a mini-grid; and using stand-alone options.

The underlying methodology used in grid extension is the Kruskal's Algorithm [2,42].

Reference electrification model

The Reference Electrification Model (REM) is developed by the Comillas Universal Energy Access Laboratory (UEA Lab) of Massachusetts Institute of Technology (MIT), USA in collaboration with IIT-Comillas in Madrid, Spain. The tool builds on a previous model called Reference Network Model which is a large scale electricity distribution planning tool [60] used by the regulators in Spain as a decision-support tool [54]. REM is designed to examine the least-cost electrification option by carrying out cost-optimization on standalone systems, mini-grids, and grid extension. The tool has a high level of spatial granularity and produces results even at the level of individual customers. For example, it is possible to specify hourly demand pattern for individual customers in REM. It prescribes infrastructure layouts for the proposed technological option as well as local generation configurations and storage options. REM has been applied in several countries such as India [61,62], Rwanda [55,63], Uganda [64], and Kenya [65] to examine different electrification planning issues ranging from small villages to country level studies. One of the key advantages of REM over other tools is that REM attempts to account for grid reliability which can play important roles in the preference for grid-extension.

Data requirement

In addition to the data required for Network Planner already listed above, REM uses data on demand for geo-located buildings (based on types of customers and demand patterns). The demand patterns are defined by hourly power consumption for both critical and non-critical demand for a year. Notably, in the standalone and mini-grid options, REM does not require data on wind and hydro. Further, REM requires information on the minimum acceptable levels of quality of supply in grid-extension and off-grid systems.

Open source spatial electrification tool (OnSSET)⁷

Open Source Spatial Electrification Tool (OnSSET) is developed by the Energy Systems Analysis Group (dESA), KTH Royal Institute of Technology, Sweden. The tool benefited from funding and support from United Nation Development Programme (UNDP), World Bank, International Energy Agency (IEA), etc. and has been applied in different countries to demonstrate its usefulness. The model is a computational optimization decision support tool that determines the least-cost electrification technology solutions to connect communities without electricity access and the amount of investment that will be required to deliver the projects that will make the electrification access connections. The objective of the model is to identify the least-cost options for generation technologies (including generation mix) and the investment that will be required to provide cost-effective solution for electricity infrastructure development to households without electricity access. The model does this by applying geospatial information to socioeconomic, demographic,

⁷ <http://www.onsset.org/> (visited on December 20, 2021).

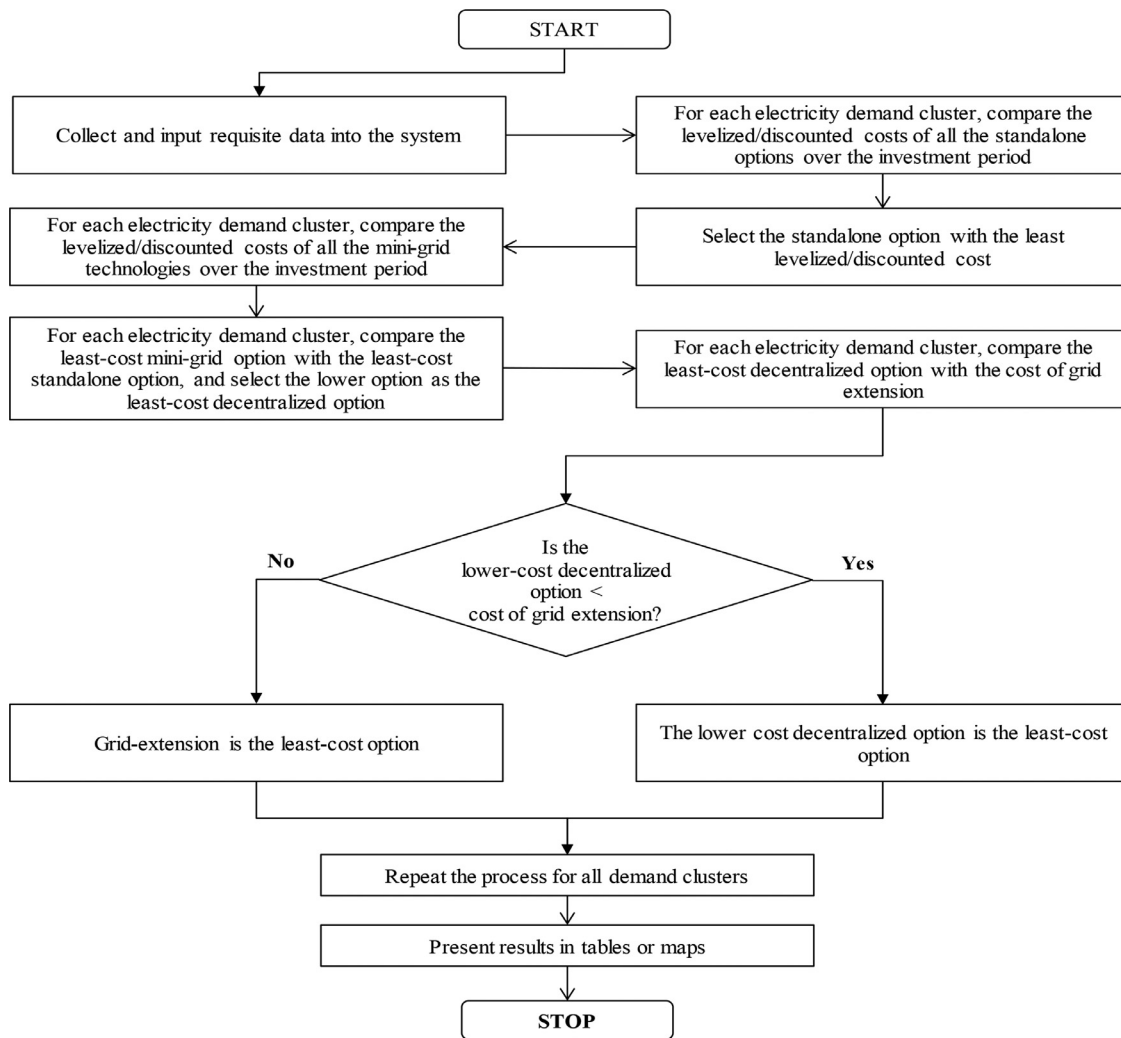


Fig. 2. Flow chart showing the decision making process for GIS-based rural electrification planning tools.

and energy demand data as well as projections using linear programming optimization techniques. Moreover, the tool is designed to support the United Nations' SDG 7: the provision of affordable, reliable, sustainable, and modern energy for all. OnSSET has been used for electrification planning in several countries and regions, including Afghanistan [48], sub-Saharan Africa [6], Nigeria [4,66], Ethiopia [9,66], Kenya [10,66], Rwanda [49], Benin [67], and Brazil [51].

Data requirement

The data required for OnSSET are similar to those for Network Planner. In addition, OnSSET also requires data on administrative boundaries of a study area, distribution transformers, road network as well as raster datasets for renewable energy potentials (for wind, solar and hydro), night time lights, land cover, digital elevation models among others. The grid-extension algorithm used by OnSSET is based on a tree search algorithm which traverses iteratively through un-electrified population settlements (which act as subsets of tree nodes) using Locality-Sensitive Hashing (LHS) to identify the nearest neighbour that may be qualified for grid-extension [49].

Comparison of the selected GIS-Based models

The objective of the models presented above is to integrate GIS in rural electrification planning in order to meet objective of increasing electricity access and promoting renewable energies. From our review,

it is possible to establish some similarities and differences amongst the models. First, the data requirements are essentially similar. The data requirement may be grouped into three broad categories (a) GIS – data (b) non-GIS data (c) Cost data [2,49,54]. The data requirements and other unique features of these tools are compared in Tables 1 & 2 respectively. Second, the underlying logic of the methodology is also very similar, even though there are differences in the actual algorithmic method adopted by each model. For example, the grid-extension algorithm used by Network Planner is the Kruskal's Minimum Spanning Tree algorithm [2,7,42] while OnSSET uses a tree search algorithm complemented by Locality-Sensitive Hashing (LHS) [49]. In the same vein, REM uses a combination of several algorithms such as optimization algorithms and heuristic methods [49]. All the models use the levelized cost of electricity generation to estimate the cost of the different technology options, before selecting the option that has the least cost. The basic underlying logic that seems to be common for these models is presented in Figure 2.

As noted by Parshall *et al* [42] electrification planning models are not meant to replace detailed engineering and architectural designs and analyses of grid and off-grid rollout (including load-flow analysis) which would be needed as part of the implementation and deployment process. Thus, such tools cannot be used as a stand-alone implementation and technology deployment tool for electrification planners and other stakeholders interested in promoting energy access to adopt.

Table 1
Comparison of the data requirement of the models.

Description	SOLARGIS	Network Planner	OnSSET	REM
Geographic data				
-Administrative boundaries (vector)			+	+
-MV, LV lines (vector)	+	+	+	+
-Existing MV/LV transformers (vector)	+	+	+	+
-Road network (vector)			+	+
-Demand clusters (vector)	+	+	+	+
-Solar energy potential (raster format)			+	+
-Micro hydropower potential (vector)			+	
-Wind potential (raster format)			+	+
-Biomass potential (raster format)				
-Night time light (raster format)			+	+
-Digital elevation model /Physical characteristics (raster format)			+	+
-Travel time (raster)			+	+
Socio-economic/demographic Data				
-Population of demand clusters		+	+	+
-Household size		+	+	+
-Population growth rate	+	+	+	+
-Inter-household distance		+		
-Electricity demand (residential),	+	+	+	+
-Electricity demand (institutional)		+	+	+
-Electricity demand (productive uses)		+	+	+
-Gross Domestic product (GDP)	+	+	+	+
-Human Development Index			+	+
-Income elasticity of electricity demand		+		

Table 2
Comparison of the models on other parameters.

Criteria	Attributes	SOLARGIS	Network Planner	OnSSET	REM
Data input platform					
	Web-based		+		
	Desktop based (works with QGIS or ArcGIS)	+		+	+
Web-based simulation					
Accessibility					
	Free		+	+	+
	Proprietary	+			+
Granularity					
	National	+	+	+	+
	Sub-national	+	+	+	+
	Local (community, clusters, demand centers)	+	+	+	+
	Buildings				+
Output					
	Maps	+	+	+	+
	Tables	+	+	+	+
Cost penalties based on terrain				+	+
Standalone options					
	Wind	+			
	Solar		+	+	
	Diesel		+	+	
Mini-grid RE options					
	Wind			+	
	Solar		+	+	
	Diesel		+	+	
	Micro-hydro			+	
	hybrid	+			

It is worth noting that the underlying methodologies in these models omit one practical aspect of rural electrification planning. Medium voltage and low voltage lines are usually constructed on existing roads and the modelling procedure in these models do not follow road networks but just the shortest linear distance to the demand node or settlements. The algorithm that simulates the cost of grid-extension needs to be such that the grid-extension follows existing or planned road network. We acknowledge that incorporating this may be difficult due to the level of programming that may be required. Furthermore, while OnSSET considers seven technology options, Network Planner covers only three technology options (i.e. solar home services, grid-extension and diesel systems) and this constitutes a setback for Network Planner.

Case studies of the models

In-dept review of the GIS-based electrification studies reveals that the application of these models have mainly been in sub-Saharan Africa.

This is mainly because countries in the region have the lowest electricity access rates and require careful planning to meet electricity access targets. In addition to the low electricity access rates, SSA countries have low infrastructure penetration and high spatial diversity in terms of renewable energy potentials. Table 1 shows a summary of where GIS-based rural electrification planning studies have been applied based on our review.

Application of OnSSET in Nigeria

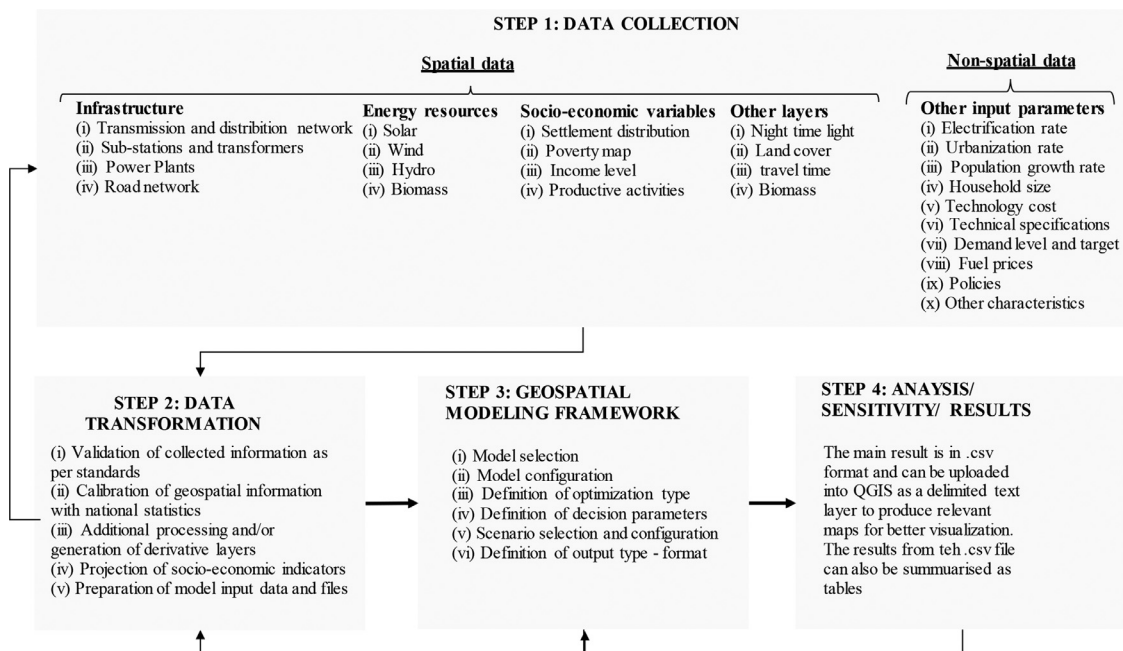
OnSSET methodology

The procedure for modelling with OnSSET is presented in Fig. 3. The first step is the process of data collected. OnSSET requires a variety of spatial data on various formats (vector or raster) as well as non-spatial data. The data are uploaded into QGIS and are validated and processed using a custom OnSSET QGIS plugin. This step produces a set of custom files in .csv format and geojson. In the next step, OnSSET python

Table 3

Summary of where GIS-based rural electrification planning studies have been applied based on our review.

Region	Country	SOLARGIS	Network Planner	OnSSET	REM	Others
Africa	Sub-Saharan Africa (multi-country)					
	Nigeria		[2,45,46]	[6] [4,66]		[53,68] [47,69,70]
	Kenya		[42]	[10]	[65]	[71]
	Tanzania			[72,73]		
	Uganda				[64]	[39,74]
	Ghana		[7]			
	Malawi			[49]		
	Liberia		[44]			
	Senegal		[43]			
	Rwanda				[55,63]	
	Tunisia	[75]				
	Ethiopia			[9,66]		
	Morocco					[35]
	Burkina Faso			[76]		[41]
	Cape Verde	[31,75]				
	Benin			[67]		
	Zambia					[77]
	South Africa					[36]
Europe	Spain	[37,38]				[27,33]
	Italy	[75]				
	Greece	[75]				
	France					[33]
South America	Brazil	[78]		[51]		[40]
	Cuba					
	Bolivia			[50]		
Asia	India	[75]			[61,62]	
	Afghanistan			[48]		
	Philippines					[79]
North America	USA					[80]

**Fig. 3.** Summary of OnSSET procedure Source: Adapted from [49].

codes are run using a Python Integrated Development Environment such as PyCharm. The .csv file produced in the previous step is uploaded to the PyCharm and the non-spatial data are also inputted. This output of this step is also a .csv file which contains many other columns showing relevant results. The .csv file may then be uploaded to QGIS again for visualization and map production. Details of the OnSSET methodol-

ogy have been provided already in the literature [49] and on OnSSET dedicated website.⁸

⁸ <https://onsset.readthedocs.io/en/latest/index.html>.



Fig. 4. Outline Map of Nigeria showing Kaduna State.

Study area

Kaduna State is one of the 36 states in Nigeria and is located in the northern part of the country between longitudes 6.093411°E and 8.99979°E and latitudes 8.785024°N and 11.49554°N. It is bordered in the north by Zamfara, Katsina, and Kano; in the east by Bauchi and Plateau; in the south by the Federal Capital Territory (Abuja) and Nasarawa; and in the west by Niger state. The map of Nigeria showing the location of the state is presented in Fig. 4.

Our objective is to use OnSSET to answer some questions related to rural electrification planning such as: what is the least-cost technology option for achieving universal electricity access in Kaduna State? What is the additional electricity generation capacity that will be required? What amount of investments will be needed? However, before we present the results of the OnSSET, it is important to present some of the underlying drivers that influence rural electrification planning in the study area.

Distribution of settlements in the study area

Data on the distribution of settlements in the study area are obtained using two sources: a .csv file containing names, coordinates, and population of settlements (disaggregated by gender) obtained from the Bill and Melinda Gates Foundation (BMGF)⁹; and a raster file of population distribution in Nigeria obtained from World Pop.¹⁰ The .csv file from BMGF is converted to an ESRI Shapefile and we buffered the point vector by 300meters to obtain a multi-polygon vector file, then we carried out an “intersection” operation with the World pop raster file. Using the shares of population in Kaduna State categorized as urban and rural (i.e. 40% urban and 60% rural). We then set a rural threshold value of 126.25 such that the resulting data of population distribution will be split in a similar ratio. We elected to adopt this approach because the

⁹ The Bill and Melinda Gates Foundation data also has population. We decided to use the data because it has names and their details of real settlements and will be more useful for policy decision. We had overlaid the bill gates data on the World Pop data to assess whether the population clusters were similar. Based on our observations, we elected the approach of buffering the data and intersecting with the World Pop data.

¹⁰ “WorldPop (www.worldpop.org) - School of Geography and Environmental Science, University of Southampton; Department of Geography and Geosciences, - Funded by The Bill and Melinda Gates Foundation (OPP1134076). <https://doi.org/10.5258/SOTON/WP00645>”. The data are in Raster format. The initial data was for the whole of Nigeria. However, we used the shapefile of the boundaries of the study area to clip it.

Table 4

Summary of distribution of settlements in Kaduna.

Description	Value
Number of settlements	10,396
Rural settlements	6,432
Urban settlements	3,964
Min. pop of settlement	10
No. of settlements having a population of 0–100	4,931
No. of settlements having a population of 101–500	4,060
No. of settlements having a population of 501–1000	838
No. of settlements having a population of Above 1000	567
Max. pop of settlement	21,584
Ave. pop of settlement	300
Base year (i.e. 2018) population based on buffered point feature	3,116,475
2018 population (Projected from 2005 Census)	6,113,503

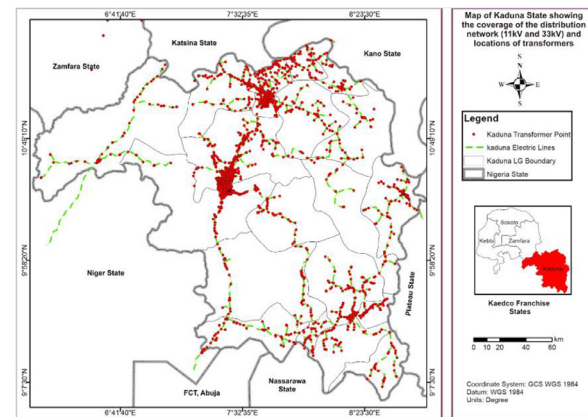


Fig. 5. Coverage of existing grid and locations of transformers Source of data: Kaduna Electricity Distribution Company (KEDCo).

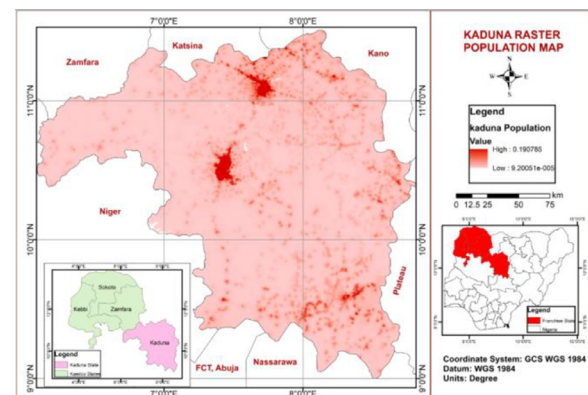


Fig. 6. Population distribution map for Kaduna State Source of data: World Pop.

resulting file will have the names and locations of actual settlements which is more useful for decision making. The summary of data on the distribution of settlements in the study area is presented in Table 4. We note that the base year population using this approach is remarkably different from the population estimate of Kaduna State using projections from the 2005 census. Notwithstanding, our approach provides granular details which is useful for decision making.

Coverage of the existing grid and electrification status of settlements

OnSSET uses data on the coverage of the existing distribution network and nighttime lights to estimate the electrification status of settlements. We present the coverage of the existing distribution network and locations of transformers in Kaduna State in Fig. 5. We also present the population map from World Pop and night-time light map in Fig. 6 and Fig. 7 respectively. We may observe the similarities in the patterns of the

Table 5
Summary of Result of OnSSET analysis in the final year 2030.

Technology Option	Number of settlements (%)	Corresponding 2030 population	Investment (million US\$)	Investment per capita (US\$/ person)	New Capacity (kW)
Mini-grid hydro	14 (0.13%)	680	0.05	70.99	11.66
Mini-grid PV	5,244 (50.44%)	1,998,303	505.08	252.76	252,532.55
Standalone PV	2,622 (25.22%)	80,960	20.56	253.98	7883.70
Grid extension	2,516 (24.20%)	1,983,482	61.65	31.08	24,616.15
TOTAL	10,396 (100%)	4,063,425	587.35		285,044.07

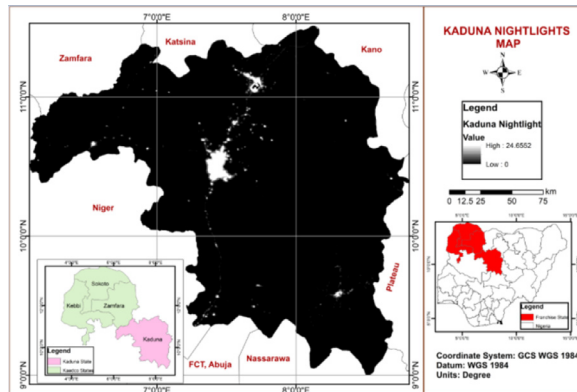


Fig. 7. Nighttime light map for Kaduna Stat.

maps in Figs. 5 to 7 which shows the correlation between places with high population density, coverage of the existing grid, and night-time lights.

Results

The summary of result showing the least-cost technology option for electrifying unelectrified settlements in Kaduna is presented in Table 5. We present only the results for the final year 2030. We observe from Table 5 that only four technology options would play role in providing electricity access in Kaduna State. Out of 10,396 unelectrified settlements, mini-grid PV is the least-cost technology for electrifying 5,244 or 50.44% of the settlements with a corresponding population of about two million. The total investment required for this is US\$505.08 million

and the new electricity generation capacity required will be 252MW. In contrast, mini-grid hydro will be the least-cost technology option for electrifying only 14 settlements. Given the location-specificity of hydro-electric power, we observe that these settlements are close to the locations of potential micro hydropower plants. The investment requirement for electrifying these 14 settlements is US\$0.05 million while the total new capacity that will be required will be 11.66 kW.

The distribution of the settlements and the least-cost technology option for electrifying them is presented in Fig 8.

Discussion

We have demonstrated how OnSSET may be used to answer rural electrification planning questions such as: what is the least-cost technology option for achieving universal electricity access in Kaduna State? What is the additional electricity generation capacity that will be required? What amount of investments will be needed? There are more specific questions that may be answered by the tool by the OnSSET tool. For example, from the results generated by OnSSET one may be able to know which technology option is appropriate for settlements with different population sizes. The number of settlements that require different sizes of new capacities of the different technologies may also be obtained from the results. These specific questions provide unique insights into decision-making for providing electricity access. With the availability of data at a granular level on more variables, planning and decision-making on electricity access will be significantly improved. One of the major challenges of GIS-based rural electrification planning is access to data at a granular level. In this way, international development agencies might play a key role by supporting national and regional institutions to collect such data and make it available for planners, policymakers and researchers on an open access basis.

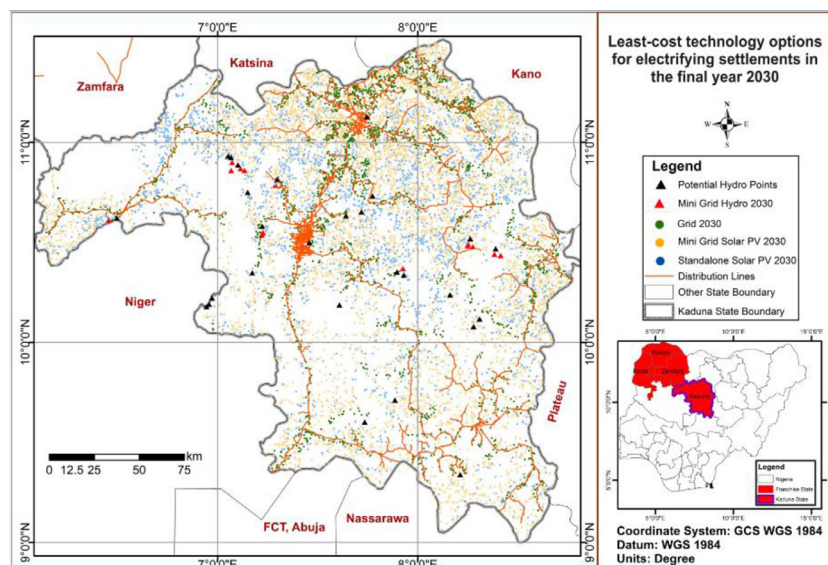


Fig. 8. Least-cost technology options for electrifying settlements in 2030.

Conclusion

This study carries out an overview of GIS-based rural electrification planning models with a trace to their historical evolutions and focuses on web-based and desktop-based models that have been developed and used extensively in rural electrification planning in recent years. Specifically, four of such models are considered: SOLARGIS, Network Planner, OnSSET, and the Reference Electrification Model (REM). These four are selected for in-depth review because of their wide use and because studies that have used them are available in English. A notable setback to this paper is that the documentation of some of the models as well as the works that employ them are in languages other than English. This makes it difficult for the authors to review these works. For each of these tools, the study examines their development, data requirements, and results. Studies that have applied each of these models are reviewed with the aim of demonstrating the applicability of the models for national and sub-national studies. Moreover, we carry out a comparison of these models and opine that these models have similar core methodological underpinnings. The newer models, i.e. OnSSET and REM have done greatly in building on the core structure laid by earlier models and have taken advantage of advancements in geo-spatial technologies and improved data availability to become more robust. In particular, we conclude that OnSSET has better capabilities than the other models. Specifically, we find OnSSET very useful because it is also open source with. Based on this, the study demonstrates the application of OnSSET in Nigeria using Kaduna State as the study area. The result shows that mini-grid PV is the least-cost technology option for electrifying about 50% of unelectrified settlements in the study area. The number of persons that will be provided access to electricity as well as the investment requirement are also generated from OnSSET. We conclude that with the availability of data at a granular level on more variables, planning and decision-making on electricity access using tools like OnSSET will be improved.

CRedit author statement

Salisu Isihak: Conceptualization, Validation, Methodology, Investigation, Resources, Writing - Original Draft

Uduak Akpan: Resources, Writing - Original Draft

Subhes Bhattacharyya: Reviewing and Editing, Supervision

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical statement

Hereby, I Salisu Rabi Isihak, consciously assure that for the manuscript “**Evolution of GIS-based Rural Electrification Planning Models and an Application of OnSSET in Nigeria**” the following is fulfilled: 1) This material is the authors' own original work, which has not been previously published elsewhere. 2) The paper is not currently being considered for publication elsewhere. 3) The paper reflects the authors' own research and analysis in a truthful and complete manner. 4) The paper properly credits the meaningful contributions of co-authors and co-researchers. 5) The results are appropriately placed in the context of prior and existing research. 6) All sources used are properly disclosed (correct citation). 7) All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content. I agree with the above statements and declare that this submission follows the policies of Solid State Ionics as outlined in the Guide for Authors and in the Ethical Statement.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.rset.2022.100019](https://doi.org/10.1016/j.rset.2022.100019)

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